

An Lam

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EDUCATION

Master of Science in Computer Science, <i>University of Texas at Arlington</i>	GPA: 4.00 - Dec 2026
Bachelor of Science in Computer Science, <i>Texas Tech University</i>	GPA: 3.96 - Dec 2022

SKILLS

Languages	JavaScript (ES6+), Python, SQL (Postgres, Oracle), C/C++, Bash, HTML5/CSS3
Full-Stack Development	React, Node.js, Express.js, RESTful APIs, Tailwind CSS, Zod, EmailJS
Data & Analytics	PostgreSQL, ETL Pipelines, Data Warehousing, Pandas, NumPy, scikit-learn, TensorFlow, Hadoop, MapReduce, Vector Space Model
Infrastructure & DevOps	Docker, Docker Compose, Nginx, GitHub Actions (CI/CD), SSL/TLS Termination, Linux Systems, Cron Scheduling
Cloud & Tools	Hetzner Cloud, Vercel, Supabase, Git, Postman, Linux/Unix Shell
Technical Focus	Full-Stack Engineering, Data Engineering, DataOps, DB Indexing & Optimization, Performance Tuning, Reverse Proxies, Production Systems, Agile

WORK EXPERIENCE

Database Administrator PostgreSQL, Python, Node.js, Selenium	Jan 2023 — Mar 2025
<i>Phuong Tran Inc</i>	<i>Mentor, Ohio</i>
- Operated a centralized PostgreSQL data warehouse, collecting POS, inventory, and scraped data via automated ETL pipelines for BI reporting. - Built scalable web scrapers using Python and Selenium to monitor 100+ competitor sites and support dynamic pricing. - Developed a full-stack scheduling system (React/Node.js), reducing technician idle time by 1+ hour daily and increasing revenue by 15%. - Optimized query performance using EXPLAIN ANALYZE and indexing to speed up reporting and analytics. - Maintained high-availability POS systems across multiple locations during peak operations.	
Student Researcher – C/C++, Bash, Linux, RISC-V	Jan 2022 — Dec 2022
<i>Texas Tech University</i>	<i>Lubbock, Texas</i>
- Engineered automated benchmarking pipelines using Bash/Makefile for large-scale RISC-V data collection. - Analyzed performance data under high-concurrency workloads to identify and resolve system bottlenecks. - Developed visualization dashboards to interpret runtime trends, accelerating technical reporting and diagnostics. - Standardized experimental workflows in Linux to ensure high data integrity for compiler research benchmarks.	

PROJECTS

Movies Big Data Lab - Live Demo	Jan 2025 — Present
<i>PostgreSQL, Node.js, React, Python, Docker Compose, Nginx, GitHub Actions, Hetzner Cloud</i>	
- Developed a full-stack web application with a React frontend, Node.js APIs, and a PostgreSQL backend to help students visualize how DBMS queries operate on data. - Built a high-performance analytics platform on PostgreSQL, managing 29M+ rows with a normalized schema optimized for large-scale analytical queries. - Achieved 20x query speedup using GIN Trigram indexes and materialized views, reducing latency to milliseconds. - Secured the platform with HTTPS/TLS, Cloudflare DNS, Netlify CDN, Nginx reverse proxying, and Supabase RLS. - Built automated CI/CD pipelines using GitHub Actions YAML and Docker for deployments on Hetzner Cloud.	
Personal Portfolio Website	Jan 2025 — Jan 2025
<i>React, Tailwind CSS, Framer Motion, EmailJS, AI-Assisted Development</i>	
- Built a high-performance personal portfolio in a 8-hour sprint using AI-assisted workflows for rapid UI/UX iteration. - Compressed 80+ hours of development into a focused, production-ready build. - Integrated EmailJS contact workflows with automated send and reply handling. - Implemented fluid UI animations using Framer Motion and Lucide React.	
Distributed Systems & Data Processing	Aug 2025 — Sep 2025
<i>C++, Bash, POSIX Threads, MapReduce, Mutex, CI/CD, Hadoop Architecture</i>	
- Engineered a custom C++ synchronization module utilizing POSIX Threads and Mutexes to manage parallel transactions and eliminate race conditions. - Architected a 2-level MapReduce engine to process 16 million records, implementing a custom shuffle-and-sort phase for optimized data distribution. - Automated data ingestion workflows using Bash/Shell scripting and integrated CI/CD pipelines to streamline testing and build reliability. - Mimicked Hadoop fault-tolerance by designing a master-worker coordination layer to ensure atomic task completion during high-volume processing.	